Securing the Sky: Lightweight Machine Learning for Zero-Day Intrusion Detection in UAV Systems*

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Abstract

Unmanned Aerial Vehicles (UAVs) are increasingly employed in civilian and commercial applications, which heightens the need for robust and reliable security mechanisms. UAV systems are exposed to a wide range of cyber and cyber-physical attacks that can disrupt missions, compromise data integrity, or lead to unsafe physical behavior. This work investigates machine learning-based intrusion detection for UAV systems with a particular focus on zero-day attack detection and computational feasibility. The objective is to assess whether lightweight classification models with low architectural complexity can achieve strong detection performance while remaining suitable for deployment in resource-constrained environments. To this end, four classifiers are evaluated. Detection performance is examined under a supervised zero-day learning framework. Experimental results demonstrate that meaningful generalization to unseen attacks can be achieved using compact machine learning models. Detection performance is competitive with results reported in related work, while computational efficiency measurements indicate promising suitability for real-world deployment.

1 Introduction

The use of Unmanned Aerial Vehicles, commonly referred to as drones, has increased significantly in recent years [22]. Once confined to military or hobbyist use, UAVs are now applied across a wide range of civil and commercial activities [7]. They support search and rescue operations, enable disaster assessment and recovery, and provide infrastructure inspection for bridges, power lines, and construction sites [16, 20]. In such scenarios, UAVs offer significant advantages due to their ability to access hazardous or hard-to-reach environments and operate with minimal human intervention. Beyond those applications, UAVs are increasingly employed in domains such as agriculture and environmental monitoring, where their mobility, operational flexibility, and cost efficiency allow large areas to be monitored [14]. These capabilities have accelerated their adoption in industrial and commercial settings and have attracted growing interest from major industry stakeholders in UAV-based systems, especially for applications such as automated goods delivery, where reliability and operational safety are of critical importance [10].

As UAVs are increasingly deployed across a wide range of application domains, their exposure to security threats likewise increases. Drones are inherently vulnerable due to their dependence on wireless communication links, unencrypted or weakly protected navigation signals, and limited onboard computational resources [6,25,36]. These characteristics expose UAV systems to a wide range of attacks, including GPS jamming and spoofing, false data injection,

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and denial-of-service, leading to severe safety and operational consequences [27]. Several UAV-targeted attacks are practically feasible in real-world settings, as commonly available tools can be exploited to launch disruptive attacks [32]. Reliable detection of such threats is, therefore, essential to ensure the secure and dependable operation of UAV systems, but it remains a significant challenge.

Intrusion detection systems represent a viable approach to enhancing the security of UAV systems, and recent years have seen an increasing adoption of machine learning-based approaches for this purpose [4,17]. However, a number of important limitations persist in existing research.

Many existing machine learning-based intrusion detection approaches rely on generic or simulation-based datasets that do not accurately reflect UAV-specific communication patterns, operational dynamics, or attack surfaces, thereby limiting their relevance for real-world deployment. Closely related to this issue, a considerable portion of prior work focuses on a narrow set of attack types, frequently restricted to GPS spoofing or jamming scenarios, which constrains the ability to assess the effectiveness of intrusion detection approaches across the diverse threat landscape faced by UAV systems. Furthermore, the majority of studies rely on evaluations conducted within a single dataset and experimental configuration. While this approach often results in high reported accuracy, it does not adequately assess whether such performance persists when models are applied under different conditions. This limits insight into the robustness and generalization of the intrusion detection approaches. Lastly, computational efficiency and detection latency are often insufficiently considered, despite their critical importance for real-time detection and potential onboard deployment in resource-constrained UAV environments.

In light of these limitations, there remains a need for the systematic evaluation of machine learning-based intrusion detection mechanisms tailored to UAV environments. In particular, an improved understanding is required regarding their robustness across datasets, their effectiveness against diverse attack types, and their suitability for deployment under the computational and latency constraints of UAV systems. This work seeks to address these challenges. The key contributions of this study can be summarized as follows:

- Design and implementation of a unified evaluation pipeline for zero-day intrusion detection in UAV systems using lightweight machine learning models, with emphasis on both detection effectiveness and runtime efficiency.
- Explicit evaluation of computational efficiency, including inference latency, to assess the feasibility of deploying the proposed classifiers under resource constraints.
- Comprehensive multi-dataset evaluation using three publicly available UAV datasets derived from real or experimental flight deployments.

2 Related Work

When looking at the broader landscape of intrusion detection systems for drones, a range of approaches has been explored. Some of the earliest works in this domain employ, for example, game-theoretic models [26,29]. Over time, however, research attention has increasingly shifted towards machine learning-enhanced intrusion detection.

When examining the learning paradigms adopted by machine learning-based intrusion detection for UAV systems, a dominance of supervised learning approaches can be observed. Most existing studies formulate intrusion detection as a classification problem and rely on labeled datasets for training and evaluation. Beyond supervised learning, prior work also explores

unsupervised anomaly detection techniques, deep learning models, and, to a lesser extent, reinforcement learning-based methods. A smaller number of studies further investigate alternative approaches that do not fit neatly into these categories; however, these approaches are represented only sporadically in the literature.

While a broad spectrum of learning-based techniques has been proposed, the effectiveness and practical relevance of these approaches are fundamentally constrained by the datasets on which they are developed and evaluated. In particular, dataset realism, UAV-specific characteristics, and the breadth of attack coverage play a decisive role in determining how meaningful the reported results are.

2.1 Datasets for Intrusion Detection

A key component in the development of effective intrusion detection mechanisms for UAV environments is the choice of the dataset used for training and evaluation. The dataset largely determines how representative the evaluation is of real-world scenarios and to what extent the resulting models can generalize beyond the experimental setup. Based on the existing literature, the datasets employed for UAV intrusion detection can be broadly grouped into three categories:

(I) generic, non-UAV-specific intrusion detection datasets,

(II) simulated or synthetically generated UAV data, and

(III) datasets explicitly collected from real or experimental UAV deployments

Furthermore, a comprehensive assessment of intrusion detection mechanisms ideally requires evaluating model behavior under varying data distributions and operating conditions, rather than relying on a single experimental setup. However, a review of recent UAV intrusion detection studies shows that most proposed approaches are evaluated on only one dataset, which limits insights into their robustness. As noted by Islam et al. [17], reported detection performance is strongly influenced by dataset selection, and results obtained under a single evaluation setting may not be reliable when models are applied under different conditions. This dependence on dataset-specific characteristics also complicates meaningful comparisons across studies.

2.1.1 Generic Intrusion Detection Datasets

Several generic intrusion detection datasets remain popular not only in traditional network security research but also in UAV-related Intrusion Detection System (IDS) studies. Frequently used datasets include KDDCup99 [1] and its refined version NSL-KDD [30], as well as more recent datasets such as CIC-IDS2017 [23]. These datasets capture a variety of network traffic patterns under normal and malicious conditions. Despite their widespread adoption, these datasets were not designed with UAV systems in mind and therefore lack UAV-specific characteristics. In the context of UAV networks, this includes the absence of flight telemetry and control commands, as well as the lack of data generated by UAV-specific communication protocols such as MAVLink. These protocols play a central role in UAV operation, enabling navigation, mission control, and sensor data exchange. As a result, models trained exclusively on generic datasets may learn patterns that are only weakly related to the operational behavior of real UAV systems.

This mismatch raises concerns regarding the applicability and generalization of such models when deployed in realistic UAV environments. While generic datasets can be useful for preliminary evaluation or for benchmarking baseline detection performance, their limited representativeness means that the high detection accuracy reported on these datasets does not

necessarily translate to effective intrusion detection on real drones or UAV networks. Consequently, an increasing number of studies have begun to explore UAV-specific datasets.

2.1.2 Simulation-based UAV Intrusion Detection Datasets

In terms of simulated versus real-flight UAV data, a clear trend can be observed in UAV intrusion detection literature: only a minority of published studies rely on data collected from real or experimental UAV deployments. This observation is consistently highlighted in recent surveys, including those by Al-Syouf et al. [4] and Ismal et al. [17], which report that the majority of evaluated approaches are validated using simulated or synthetically generated datasets. In particular, Al-Syouf et al. note that approximately 70 % of the studies they included in their review rely primarily on simulation-based data. Many of those simulation-based datasets exist only within individual studies and are not available for benchmarking or cross-study evaluation.

The use of simulated data offers several advantages for research. Simulation environments provide fine-grained control over experimental conditions and allow the systematic exploration of attack scenarios that may be costly, unsafe, or legally restricted to reproduce in real deployment. Moreover, simulations enable the generation of large, labeled datasets, which is particularly beneficial for supervised learning approaches that require extensive training data.

However, reliance on simulated data also introduces limitations. Simulation models simplify aspects of UAV dynamics, sensor noise, and environmental interactions. As a result, simulated datasets often exhibit cleaner, more regular patterns than those observed in real-flight data. Models trained on simulated data might therefore generalize only to a limited extent when deployed in real UAV systems, especially when evaluated solely within the same simulation framework.

2.1.3 Real-Flight UAV Intrusion Detection Datasets

In response to the limitations of generic and simulation-based data, recent research has increasingly focused on the creation of real-fight UAV intrusion detection datasets. However, the availability of such datasets, particularly in the form of publicly accessible benchmark datasets, remains limited. A recent survey of UAV intrusion detection datasets by Mohammed and Fourati [19] identifies nine datasets targeting UAVs and their communication systems; however, approximately half of these are simulation-based or focus primarily on anomaly and fault detection rather than on explicit attack scenarios.

2.1.4 Attack Coverage

The scope of attack types evaluated in intrusion detection studies is closely tied to the choice of dataset. As discussed previously, generic network intrusion datasets such as KDDCup99 and CIC-IDS2017 typically include multiple categories of attacks, enabling evaluation across diverse threat scenarios. Likewise, most UAV-specific datasets derived from real deployments, including those employed in this study, cover two or more distinct attack types in order to reflect the diverse threat landscape faced by UAV systems.

In contrast, studies relying on simulated data frequently concentrate on a single attack scenario, and datasets are often constructed to model one specific threat. For example, Slimane et al. [28] evaluate boosting- and bagging-based models for the detection of jamming attacks. Their dataset is generated using a Simulink-based model that simulates communication between the UAV and its ground control station under benign and jamming conditions. Similarly, Agyapong et al. [3] employ deep learning models to detect GPS spoofing attacks using data

generated in the Gazebo simulation environment. While such single-attack studies provide valuable insights into the detectability of individual threats, they offer a limited view of intrusion detection performance in more complex environments. In real-world UAV operations, multiple attack types may occur. Given that a substantial portion of existing UAV IDS research relies on simulation-based datasets, this tendency towards single-attack evaluation is quite widespread in the literature.

2.2 Comparison with Closest Related Work

Taken together, the existing literature demonstrates a broad range of machine learning approaches for UAV intrusion detection, along with substantial variation in dataset choice and attack coverage. To enable a more focused and meaningful comparison with related work, the following overview narrows the scope to studies that evaluate intrusion detection models using UAV-specific datasets derived from real-flight experiments rather than purely from simulation environments.

Ref.	Model / Algorithms	Performance Metrics	Dataset Used
[33]	One-Class Support Vector Machine, Autoencoder Neural Network, Local Outlier Factor	F1-Score of 81.17%, 94.81%, and 58.93% respectively	UAV Attack
[18]	Feedforward convolutional neural network	Accuracy of 98.23%	UAV-IDS
[31]	Long Short-Term Memory (LSTM)	Accuracy of 98.9%	UAVCAN, UAV Attacks, Kitsune Network Attack
[8]	XGBoost	Accuracy of up to 99.95%	Cyber-Physical
[15]	Support Vector Machine, Feedforward neural network (FNN), LSTM, Convolutional Neural Network (1D-CNN)	F1-Score of 89.86%, 91.9%, 94.88%, and 96.13% respectively	Cyber-Physical
[12]	One Class Support Vector Machine, Isolation Forest, Local Outlier Factor, Denoising Autoencoder, LSTM Autoencoder, Convolutional Autoencoder	F1-Score of 61.97%, 78.96%, 55.43%, 96.38%, 98.33%, and 98.51% respectively	UAV Attack

Ref.	Model / Algorithms	Performance Metrics	Dataset Used
[2]	Convolutional Neural Network	Accuracy of 98.59% and F1-Score of 98.69%	UAV-IDS
[24]	Autoencoder	NA	MAVLINK
[13]	Isolation Forest, Multi-Layer Perceptron, FNN, Model-Agnostic Meta-Learning	F1-Score of 84.61%, 90.40%, 99.67%, and 96.20% respectively (best experiment)	UAV-NIDD
[5]	Deep Neural Network	Accuracy of 99.99%	UAV-IDS
[9]	One Class Support Vector Machine, Isolation Forest, Local Outlier Factor, Stacked Autoencoder	F1-Score of 77.8%, 83.6%, 87.1%, and 92.9% respectively	UAV Attack
[21]	Random Forest, K-Nearest Neighbors, Support Vector Machine, Logistic Regression, Naive Bayes, Decision Tree, Multi-Layer Perceptron	F1-Score of 90%, 86%, 89%, 90%, 91%, 92%, 89% respectively	GPS Spoofing

Table 1: Summary of related work.

Building on the challenges identified throughout this section, several recurring limitations can be observed in existing UAV intrusion detection research. These include heavy reliance on generic, simulation-based and single datasets, narrow attack coverage, and insufficient consideration of computational efficiency under UAV-specific constraints. Considerations regarding computational efficiency are non-trivial, independent of the IDS deployment location within the UAV system.

While not inherently a limitation, prior studies differ substantially in their evaluation objectives, especially with respect to whether models are assessed exclusively on known attack types or also on attacks not encountered during training. Considering the diverse threat landscape UAV systems face, zero-day detection represents an important research direction. Accordingly, prior work has explored zero-day detection primarily through unsupervised learning approaches, but also within supervised learning frameworks, where models are evaluated based on their ability to generalize from limited attack knowledge to previously unseen attack classes.

To better position the contributions of this work within the existing literature, Table 2 provides a structured comparison of the representative studies presented above. The comparison considers four key evaluation dimensions:

1. whether detection of unseen attacks is explicitly evaluated,
2. whether multi-dataset evaluation is performed,
3. whether multiple attack types are covered, and

4. whether computational efficiency or resource-related performance metrics are reported

Entries marked with "(X)" in the first column of Table 2 indicate that the authors describe their proposed method as capable of detecting attacks not encountered during training, but do not provide explicit experimental validation for it.

Reference		**Unseen Attacks Evaluation**	**Multi-Dataset Evaluation**	**Multiple Attacks Coverage**	**Computational Efficiency Reporting**
Whelan et al.	[33]	X			X
Hadi et al.	[18]	(X)		X	
Tlili et al.	[31]	(X)	X	X	
Cordeiro et al.	[8]	(X)		X	
Hassler et al.	[15]	X		X	
Fraser et al.	[12]	X			
Al-Haija and Al Basawi	[2]	(X)		X	X
Park et al.	[24]	X		X	
Hadi et al.	[13]	X		X	
Alheeti et al.	[5]	X			
Da Silva et al.	[9]	X		X	
Nayfeh et al.	[21]				X
This work		X	X	X	X

Table 2: Overview of related work with respect to unseen-attack detection, dataset diversity, and computational efficiency.

3 Methodology

Three publicly available UAV datasets from real-flight deployments were selected, each capturing complementary aspects of UAV operation, communication, and attack behavior:

- Unmanned Aerial Vehicles Network Intrusion Detection Dataset (UAV-NIDD) [13],
- Cyber-Physical Dataset for UAVs Under Normal Operations and Cyberattacks [15],
- UAV Attack Dataset [34]

Although the datasets differ in terms of data modality and abstraction level, none of them represent generic network or sensor datasets independent of UAV operation. Each dataset is collected in the context of active UAV flights and is therefore inherently tied to UAV behavior, either through onboard telemetry or through communication traffic associated with UAV command and control. While the network data in the Cyber-Physical dataset relies on standard

wireless protocols, its tight coupling with flight telemetry ensures that it reflects realistic UAV behavior along with the other two datasets.

The chosen datasets enable the evaluation of the proposed approach across multiple experimental setups, attack vectors and deployment perspectives. For example, the datasets cover complementary parts of the UAV attack surface, including attacks targeting the UAV itself as well as wireless communication links between the UAV and external components like the ground control station. The datasets also map to distinct intrusion detection placements: the UAV Attack dataset provides onboard host-based telemetry, the UAV-NIDD dataset offers network-level traffic from wireless links, and the Cyber-Physical dataset combines both modalities for ground-control analysis.

In order to ensure a consistent evaluation across datasets, a unified experimental pipeline was applied throughout this study. Each of the four classifiers was trained and evaluated on all three datasets using an identical sequence of processing steps wherever possible. Dataset-specific preprocessing steps were limited to initial data alignment and consolidation, as well as data cleaning, after which a common pipeline was used for feature selection, model training, and evaluation.

3.1 Models

Among modern machine learning approaches for tabular data, ensemble methods, and in particular gradient boosted machines, have demonstrated consistently strong performance across a wide range of applications. Two widely adopted gradient boosting frameworks, XGBoost and LightGBM, were therefore included in this study, as both have shown near state-of-the-art performance on structured data [11, 35]. Both were configured with relatively shallow trees and a limited number of estimators, ranging from 50 to at most 150 trees depending on the experimental setting, to maintain low model complexity.

Alongside the boosting-based models, a Random Forest classifier was included as a classical and well-established baseline ensemble method, also constrained to a comparable tree count to ensure a fair comparison across architectures.

Finally, to provide a neural network-based comparison point, a compact MLP was implemented. The Multi-Layer Perceptron (MLP) was intentionally kept shallow and lightweight to allow for an assessment of how a simple neural model compares to tree-based ensembles under resource constraints, with a minimal network structure consisting of a single hidden layer. No deep or computationally intensive architecture was pursued to align with the goal of deployability in resource-constrained environments.

Hyperparameter tuning was performed conservatively using a predefined grid search to avoid over-specializing the model to the specific attack type used during training.

3.2 Pre-processing

Data preparation and preprocessing were performed prior to model training and comprised several stages. Label filtering was applied to the UAV-NIDD and Cyber-Physical datasets to ensure stable training and consistent feature representation across classes. In the UAV-NIDD dataset, the reconnaissance attack class was excluded due to having only 6 samples compared to 193,171 in the largest class, which was insufficient for learning a representative attack distribution. For the Cyber-Physical dataset, the false data injection class was removed because its substantially different feature set would have significantly reduced the common feature subset available to all remaining classes. Because the UAV Attack and Cyber-Physical datasets contained streams sampled at different rates, a merging and alignment step was required. A unified

timeline was created from the union of all timestamps, and missing values were forward-filled. Segments of the merged dataframe lacking overlapping streams were then trimmed. Finally, missing or corrupted entries were removed, and the remaining labels were encoded in a binary format (attack/no-attack).

The resulting datasets include a diverse range of threats, including Denial-of-Service (DoS), Man-in-the-Middle (MITM), jamming, replay, and spoofing attacks, among others.

3.3 Feature Selection

For feature selection, filter-based methods were employed due to their computational efficiency and interpretability. A variance-threshold filter removed non-informative features exhibiting low or zero variance, as such features contribute no discriminative power for classification. Pearson correlation, which quantifies the linear relationship between two variables, was applied to eliminate redundancy. Variables with a correlation coefficient exceeding 0.85 were identified as highly correlated, and among each correlated pair, the feature with the lower variance was discarded.

3.4 Training and Evaluation

Although commonly associated with unsupervised learning, zero-day detection can also be achieved within a supervised learning framework by leveraging a model's ability to generalize beyond the specific attack patterns observed during training (e.g., [13, 15]). In this leave-one-attack-out setting, the model is trained using labeled data but evaluated on attack types that were not included during training.

A fixed 80/20 split was first applied to the benign samples to obtain a benign portion for the training and test set. The benign split is performed once and kept consistent across all experiments to ensure comparability between training scenarios. Models are trained on the training portion of the benign samples and all samples of one attack class A_i. Then, they are tested on the benign test samples and all samples of the attack classes that have not been used for model training ($A_j, j \neq i$). This process is repeated until every attack class has been used as training attack once.

3.5 Further Feature Reduction

In addition to training models using the feature set obtained after variance- and correlation-based filtering, further experiments were conducted using further reduced feature subsets. This was achieved using mutual information (MI), which quantifies the statistical dependence between two variables; in this case, an individual feature and the class label. By computing mutual information scores, the remaining features were ranked by how strongly they are associated to the target variable. Then, training was repeated using increasingly smaller subsets of the k highest ranked features, with k ranging from small subsets to the full feature set ($k \in \{8, ...,$ full set$\}$)..

Mutual information scores were computed exclusively on the training data, consisting of the benign training samples and all samples of the attack class used for training to ensure that feature relevance estimation remains strictly confined to the information available during training and avoid leakage from unseen attack classes. For each value of k, models were trained and evaluated using the same evaluation protocol described perviously.

The goal of these experiments was to progressively reduce input dimensionality while evaluating the extent to which zero-day detection performance could be maintained. Reducing the

number of required features has practical relevance in resource-constrained UAV environments, as it reduces the dimensionality of the input representation, limits dependency on a large number of features, and supports the design of simpler, lower-complexity, and more deployable intrusion detection models.

4 Results

The UAV-NIDD dataset was selected as the primary evaluation dataset and was used for in-depth detection performance analysis, feature subset experiments based on mutual information ranking, and computational efficiency assessment. To complete this detailed evaluation, the Cyber-Physical dataset and UAV Attack dataset were employed to examine the robustness and consistency of model behavior across different data sources.

Results and confusion matrices are shown for a representative run with random seed 42. To assess the impact of model stochasticity, all experiments were repeated with five different random seeds. Across these runs, the mean and standard deviation of performance metrics were computed while keeping the data splits fixed.

Overall, the observed variance across random initializations was low for the majority of attack types, particularly for the tree-based ensemble models. In most cases, standard deviations remained within a few percentage points, indicating that the reported results are robust and not dependent on a specific random seed. Importantly, the relative performance trends across attacks and models remained consistent across all runs.

4.1 Baseline Detection Performance with Full Feature Set

Firstly, baseline classification performance was evaluated using the full feature set. This established a reference point against which the impact of subsequent feature reduction could be assessed.

Table 3 summarizes detection performance for each model under the training attack that yielded the strongest generalization to unseen attacks. Precision, recall, and F1-score are reported for the attack class.

Model	**Seen Attack**	**Recall**	**Precision**	**F1-score**	**Accuracy**
Random Forest	MITM	0.8465	1.00000	0.9169	0.8473
XGBoost	Fake Landing	0.9715	0.99996	0.9855	0.9716
LightGBM	Jamming	0.9785	0.99997	0.9891	0.9786
MLP	Fake Landing	0.9214	0.99993	0.9591	0.9217

Table 3: Baseline detection performance for the best-performing training attack using the full feature set.

Precision values for the attack class reached 1.00 or close to 1.00 across all models. This behavior is largely due to the extreme class imbalance in the test sets, where attack samples outnumber benign samples by several orders of magnitude.

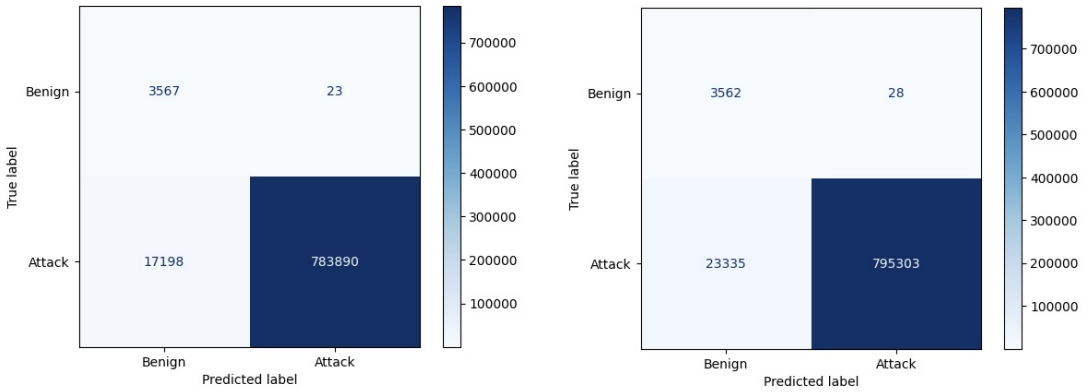


Figure 1: Confusion matrices for LightGBM trained on the jamming attack (left) and XGBoost trained on the fake landing attack (right) when evaluated on unseen attacks.

Figure 1 shows the confusion matrix for the best-performing model, LightGBM, when trained on the attack that yields the best detection performance. Consistent with the aggregate metrics, LightGBM exhibited a substantially lower number of false negatives compared to the Random Forest classifier and the Multi-Layer Perceptron, as shown in Table 2. Among the evaluated models, LightGBM achieved the highest attack F1-score (0.979). XGBoost, the second boosting-based model, attained a slightly lower attack F1-score (0.972). For comparison, Figure 1 also shows the corresponding confusion matrix for XGBoost. For both models, XGBoost and LightGBM, a non-negligible number of false negatives remained, indicating that some attack samples remain difficult to detect.

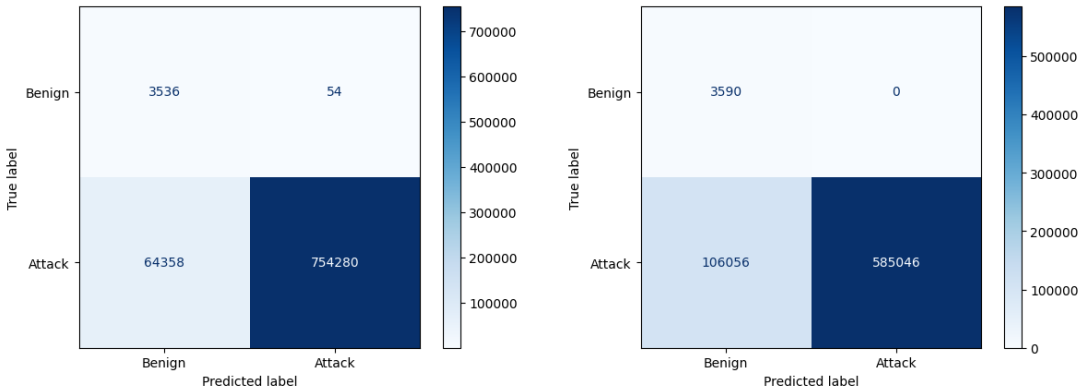


Figure 2: Confusion matrices for the MLP trained on the fake landing attack (left) and Random Forest trained on the MITM attack (right) when evaluated on unseen attacks.

When comparing confusion matrices, it should be noted that while the number of benign test samples is identical across models, the number of attack test samples varies depending on which attack is used for training, and thus absolute false negative counts should be interpreted in the context of the corresponding test size.

4.1.1 Impact of Training Attack Choice

Detection performance is not determined by model choice alone, but also by the attack used for training. Figure 3 summarizes attack class F1-scores across all evaluated models with respect to the attack used for training, with attacks sorted by mean performance in descending order.

The heatmap reveals several patterns. First, certain attacks, such as man-in-the-middle and fake landing, enabled strong generalization to unseen attacks across all models. Other attacks show a more model-dependent behavior. For example, jamming yielded near perfect performance for the boosting-based models while resulting in substantially lower performance for the Multi-Layer Perceptron, and poor performance for the Random Forest classifier. Similar, though less pronounced, patterns are observed for the scanning attack. Among all models, the Multi-Layer Perceptron displayed the most stable performance across attacks, with fewer extreme performance variations and a smaller number of training attacks resulting in F1-scores below 0.8.

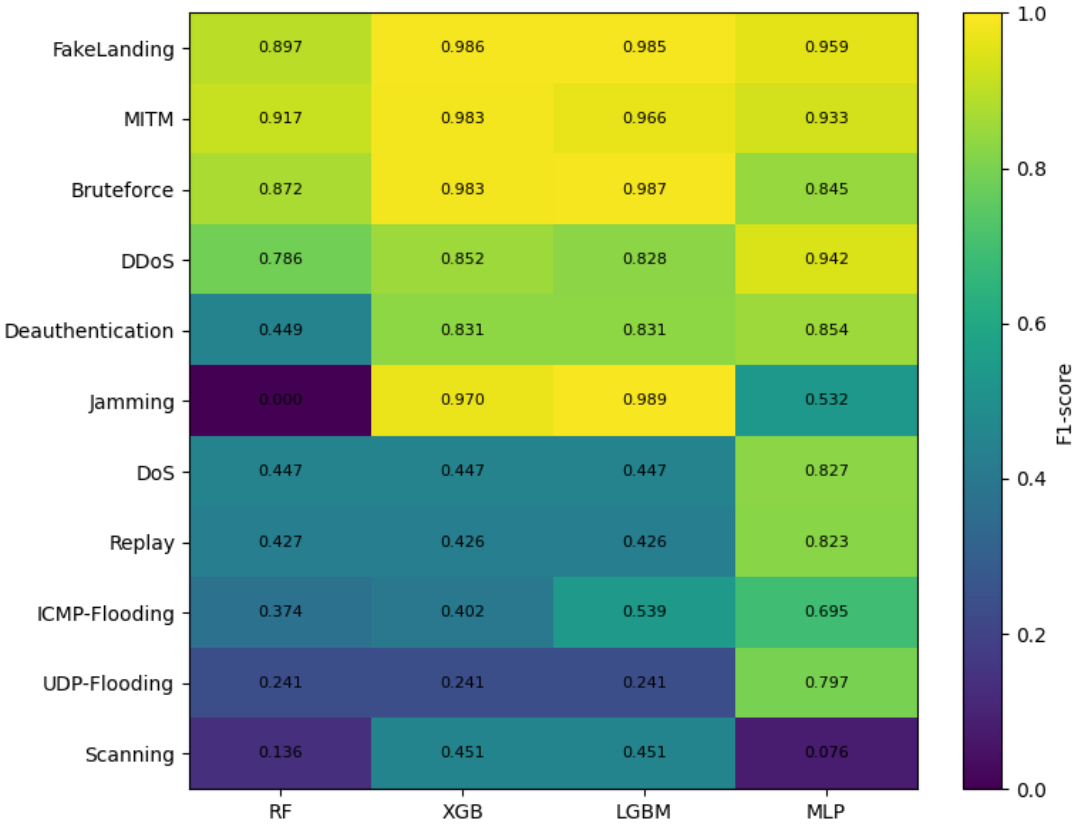


Figure 3: Detection performance (attack class F1-score) for each model as a function of the attack used for training.

4.2 Detection Performance on Feature Subsets

To assess whether further feature reduction is possible without degrading detection performance, experiments were repeated using subsets of the top k features ranked based on mutual information.

Table 4 shows the results of these subset experiments. The best detection performance achieved by each model is presented, along with the corresponding training attack and the number of retained features.

Model	**Seen Attack**	**k**	**F1-score**	**Recall**
Random Forest	MITM	15	0.9266	0.8632
XGBoost	Jamming	8	0.9935	0.9872
LightGBM	Jamming	18	0.9891	0.9785
MLP	Fake Landing	14	0.9970	0.9941

Table 4: Best detection performance after MI-based feature selection for each model, showing the training attack along with the number of retained features k that yield the highest F1-score.

After feature reduction, the Multi-Layer Perceptron achieved the highest overall detection performance, outperforming both XGBoost and LightGBM. XGBoost reached near-optimal performance with the smallest feature subset.

Compared to Hadi et al. [13], who report an F1-score of 0.9718 on the same dataset using a feed-forward neural network, this best model achieves an F1-score of 0.9970.

4.3 Evaluation of Computational Efficiency

Computational efficiency was evaluated for the model instance that achieved the strongest detection performance. Inference latency and throughput were therefore measured for the corresponding training attacks reported in Table 3.

All classifiers were assessed using the full feature set from the standard preprocessing pipeline to ensure fair comparison across architectures. Measurements were conducted on an Apple M3 CPU system after a warm-up phase, capturing both single-sample latency (batch size 1, approximating online detection) and batch latency (batch size 1024, reflecting vectorized execution). Reported values reflect model inference time only, excluding feature extraction and data acquisition, thus representing lower-bound estimates for real-world deployment. Results are summarized in Table 5.

Although computational efficiency was evaluated only on the UAV-NIDD dataset, the resulting measurements provide a representative estimate across all three datasets. The evaluated models share comparable architectures and operate on dense numerical feature vectors with similar dimensionality and identical preprocessing pipelines.

Overall, the tree-based boosting models as well as the MLP exhibited substantially lower inference latency than the Random Forest classifier. The Multi-Layer Perceptron achieved the lowest latency and highest throughput, making it the most computationally efficient model. XGBoost and LightGBM also demonstrated strong efficiency, with inference times well below one millisecond per sample. In contrast, the Random Forest classifier showed significantly higher single-sample latency.

Model	Batch (μs/sample)	Single-sample (μs/sample)	Throughput (samples/s)
Random Forest	13.68	14564.56	73,095
XGBoost	0.32	188.63	3,102,638
LightGBM	0.74	411.42	1,343,612
MLP	0.18	44.83	5,612,370

Table 5: Comparison of per-sample inference latency (measured in batch mode and single-sample mode) and throughput.

4.4 Cross-Dataset Evaluation

Model generalization was tested on the Cyber-Physical and UAV Attack datasets to determine whether performance transfers across different UAV platforms, data modalities, and attack scenarios.

4.4.1 Cyber-Physical Dataset

Table 6 summarizes the results for each model. Precision, recall, and F1-score are reported for the attack class.

Model	Seen Attack	Recall	Precision	F1-score	Accuracy
Random Forest	DoS	0.512	0.999	0.677	0.580
	EvilTwin	0.000	0.000	0.000	0.098
	Replay	0.607	1.000	0.755	0.663
XGBoost	DoS	0.409	0.984	0.578	0.486
	EvilTwin	0.157	1.000	0.271	0.239
	Replay	0.845	0.999	0.916	0.867
LightGBM	DoS	0.528	0.988	0.688	0.588
	EvilTwin	0.158	1.000	0.273	0.240
	Replay	0.844	0.998	0.915	0.865
MLP	DoS	0.445	0.990	0.614	0.519
	EvilTwin	0.028	0.993	0.055	0.123
	Replay	0.859	0.996	0.923	0.877

Table 6: Cyber-Physical dataset detection performance for all training attacks.

Overall, the Cyber-Physical dataset exhibited less variability in detection performance with respect to the choice of training attack compared to the UAV-NIDD dataset. As shown in Table 6, all models consistently achieved their best performance when trained on the replay attack, while the same training attack (evil twin) also yielded the weakest performance for all models. In contrast to the UAV-NIDD dataset, differences between the boosting models and

the Multi-Layer Perceptron were minimal.

Among the evaluated models, the MLP achieved the highest performance, with an F1-score of 0.923. The corresponding confusion matrix is shown in Figure 4. The overall detection performance remained noticeably lower than that achieved on the UAV-NIDD dataset.

In comparison, Hassler et al. [15] report that their best-performing 1D-CNN achieved an F1-score of 0.803 under comparable single-attack training conditions. Notably, they report improved performance (up to 0.961 F1-score) when training on multiple attacks simultaneously, suggesting multi-attack training could further enhance generalization.

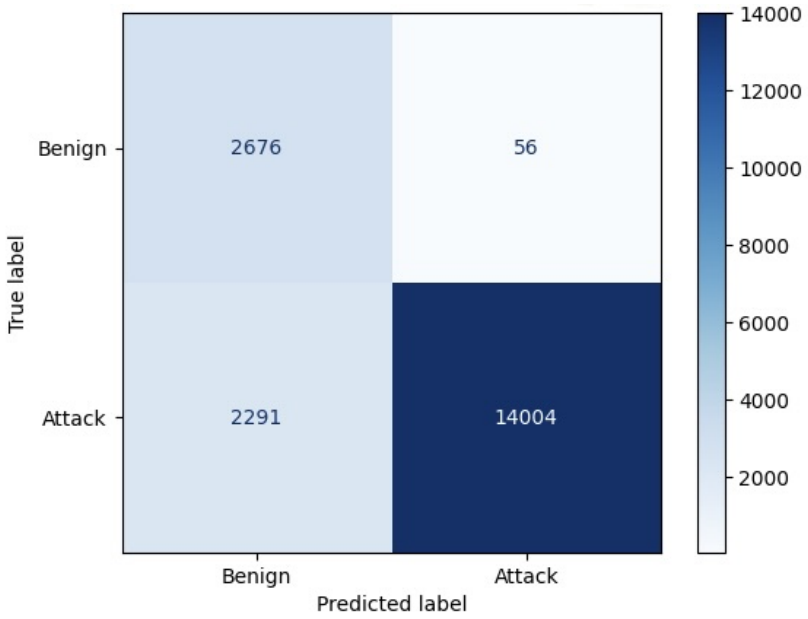


Figure 4: Confusion matrix of the Multi-Layer Perceptron for the best-performing training attack (Replay).

4.4.2 UAV Attack Dataset

Table 7 presents the detection performance of all evaluated classifiers. Precision, recall, and F1-score are reported for the attack class.

Across all models, perfect performance was observed when trained using the spoofing attack, with all classifiers achieving an accuracy and F1-score of 1.00. In contrast, performance when trained on jamming exhibited noticeably larger variability across model architectures, ranging from strong detection performance (e.g., XGBoost) to substantially weaker results (e.g., LightGBM and MLP).

Although these results demonstrate strong empirical performance, the perfect spoofing detection across diverse classifiers suggests that the dataset may contain features that strongly separate benign and attack conditions, potentially inflating performance metrics. Given the limited number of attack classes in the dataset, it remains unclear whether these findings generalize to other attack types beyond those considered.

For this same dataset, Da Silva et al. [9] report a maximum F1-score of 0.929 using a

Model	Seen Attack	Recall	Precision	F1-score	Accuracy
Random Forest	Jamming	0.7829	1.00	0.8782	0.8380
	Spoofing	1.0000	1.00	1.0000	1.0000
XGBoost	Jamming	0.9709	1.00	0.9852	0.9783
	Spoofing	1.0000	1.00	1.0000	1.0000
LightGBM	Jamming	0.0000	0.00	0.0000	0.2536
	Spoofing	1.0000	1.00	1.0000	1.0000
MLP	Jamming	0.1523	1.00	0.2640	0.3662
	Spoofing	1.0000	1.00	1.0000	1.0000

Table 7: UAV Attack dataset detection performance for all training attacks.

stacked autoencoder, treating both spoofing and jamming as a single attack class. Whelan et al. [33] report maximum F1-scores of 0.841 for spoofing and 0.914 for jamming using one-class classifiers.

5 Conclusion and Future Work

Overall, this work demonstrates that effective zero-day intrusion detection for UAV systems can be achieved using compact, low-complexity machine learning models, and that high accuracy can be attained alongside computational efficiency. Across all three datasets, several classifiers achieved meaningful generalization to unseen attacks, indicating that learning from a single known attack can yield transferable detection behavior. Performance varied by dataset—strongest on UAV-NIDD and UAV Attack, somewhat reduced but competitive on Cyber-Physical—reflecting differences in data modalities rather than fundamental limitations of the approach. Inference latency measurements confirmed that three of four models (MLP, XGBoost, LightGBM) operated well below one microsecond per sample under CPU-only execution, supporting feasibility for resource-constrained deployment.

Despite the promising results, some limitations should be acknowledged. First, the evaluation was conducted using offline datasets, and real-time deployment aspects, such as data acquisition, feature extraction latency, and other factors contributing to computational overhead, were not implemented in a live system. Although model inference latency was shown to be minimal, these additional components may introduce non-negligible runtime costs. Furthermore, all computational efficiency measurements were performed on relatively high-performance desktop hardware. Inference latency on resource-constrained UAV onboard hardware may therefore differ.

A further limitation concerns the availability and diversity of publicly accessible UAV intrusion detection datasets. Although three complementary datasets were evaluated in this work, two of them contain only a limited number of attack scenarios, which constrains the extent to which conclusions can be generalized to the broader UAV threat landscape.

Lastly, an even more rigorous evaluation strategy would involve cross-dataset testing, in which a model is trained on one dataset and evaluated on a separate dataset. Such an approach could further mitigate the influence of dataset-specific artifacts and provide stronger evidence regarding real-world deployment robustness.

5.1 Future Directions

This work points to several directions for future research. Incorporating multi-attack strategies is a natural direction for future work, as exposure to multiple attack types is intuitively expected to improve the detection of unseen attacks and has also been supported by prior work. In particular, such strategies may help reduce the remaining false negatives observed for several models.

From a systems perspective, implementing the proposed intrusion detection pipeline in a real-time UAV or ground-control environment would allow for end-to-end performance evaluation, including feature extraction, synchronization, and other computational overhead.

Beyond extensions of the presented work, future research would benefit from the establishment of more standardized evaluation benchmarks. In particular, the inclusion of computational efficiency metrics, such as inference latency, throughput, and resource consumption, would enable more meaningful comparisons between proposed methods. In parallel, continued efforts to improve dataset quality and diversity are essential to support robust model development and realistic experimental evaluation.

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